

MA 403 Real Analysis

Homework 11

1. Let $f : (0, 1] \rightarrow \mathbb{R}$ be a function, and suppose that f is Riemann-integrable on $[c, 1]$ for each $c > 0$. Recall the definition of the improper integral:

$$\int_0^1 f(x)dx := \lim_{c \rightarrow 0} \int_c^1 f(x)dx$$

if this limit exists and is finite.

- (a) If f is Riemann-integrable on $[0, 1]$, show that this definition agrees with the old one.
- (b) Construct a function f such that the above limit exists, but it fails to exist with $|f|$ in place of f .
2. Let $\gamma_1 : [a, b] \rightarrow \mathbb{R}^k$ be a path. Let $\phi : [c, d] \rightarrow [a, b]$ be a continuous, 1-1, onto map such that $\phi(c) = a$. Define $\phi_2(s) = \gamma_1(\phi(s))$.
- (a) Prove that γ_2 is a rectifiable curve if and only if γ_1 is a rectifiable curve.
- (b) Prove that γ_2 and γ_1 have the same length.
3. For any two real sequences $\{a_n\}$ and $\{b_n\}$ which are bounded below, show that

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n.$$

4. Let $\{a_n\}$ be a sequence of real numbers. Prove that:

- (a) $\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n$.
- (b) The sequence $\{a_n\}$ converges if and only if $\limsup_{n \rightarrow \infty} a_n$ and $\liminf_{n \rightarrow \infty} a_n$ are both finite and equal, in which case, $\lim_{n \rightarrow \infty} a_n = \limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n$.

5. Assume that $\{a_n\}$ and $\{b_n\}$ are two sequences such that $a_n \leq b_n$ for each n . Prove that $\limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} b_n$ and $\liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} b_n$.

6. In each case, test for convergence of $\sum_{n=1}^{\infty} a_n$.

(a) $a_n = \sqrt{n+1} - \sqrt{n}$ (b) $a_n = \frac{\sqrt{n+1} - \sqrt{n}}{n}$ (c) $a_n = (\sqrt[n]{n} - 1)^n$